S U N N A N D Æ G v1.2.0

created by k.odk

Made to be played

Sunnandæg is made to be played, a little like a modular synthesizer patch that lives in your hand. You shape a single voice by touch, by light and by the angle you hold the phone at, and listen to where it goes. It is happiest treated as one instrument among your others, rather than as a place to arrange a whole song.

At its heart is a loop. A plain sine wave is folded three times over, and the folded result is sent back to the oscillator that made it. From that moment the sound starts to answer itself: the same gesture rarely returns the same sound twice. Your job is less to command it than to lean on it, nudge it, and catch the moments you like.

Almost everything that moves inside the instrument can be sent somewhere else. That is the second idea, and most of this guide is about it: **who sends, who receives**, and how to read a patch at a glance.

Too much to learn? Skip it. Press the random keys in the SETTING panel, switch on the camera, and play with whatever comes out — wave a hand in front of the lens, tilt the phone, let the ball roll. That is a perfectly good way to use Sunnandæg. Before you press them, look at the **PROBABILITY** meter above the keys: it decides where the random pitches land. Keep its hills to the left, in the low, slow range — that is where Sunnandæg sounds best. Toward the right, the fast range tends to turn into noise.

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What it needs

Sunnandæg runs a dense, real-time synthesis engine while it watches the camera and reads the motion sensors. It asks more of the device than a typical app, so please read this page before buying or playing.

System

IOS	iOS 18.0 or later.
DEVICE	iPhone and iPad, played in landscape. On iPad the whole instrument fills the screen and every panel opens full size.
RECOMMENDED	iPhone 13 or newer. Sunnandæg is developed and tested on an iPhone 13. Older iPhones that run iOS 18 will open it, but with the camera on you may hear crackles or see the device get hot sooner.
SENSORS USED	Camera (front or back), microphone (for Micro Sound recording), and the motion sensors (tilt and the gravity ball). Each is optional — the instrument plays without them — and iOS will ask your permission the first time.
LISTENING	Headphones or good speakers. The feedback loop produces very low and very high content that small phone speakers cannot reproduce.
PRIVACY	Nothing leaves the device. The camera picture is analysed for colour, brightness and movement and is never stored or sent; the app does not use the network.

Performance and heat

The camera is the heaviest part. If the sound starts to crackle:

- Watch the **load readout** in the System / Presets panel. It shows how much of the audio deadline the engine uses: above about 70% it turns amber, and crackling begins near 100%. A letter appears when the phone warms up — F, S or C for fair, serious, critical.
- Lower the **camera frame rate** (1–30 fps). A lower rate runs cooler and reacts more coarsely.
- Turn on **Safe Mode**. When the phone gets hot it lowers the camera frame rate and widens the audio buffer automatically, and restores your settings once it cools.
- Turn the camera off entirely with its master switch when you do not need it.

About AUv3

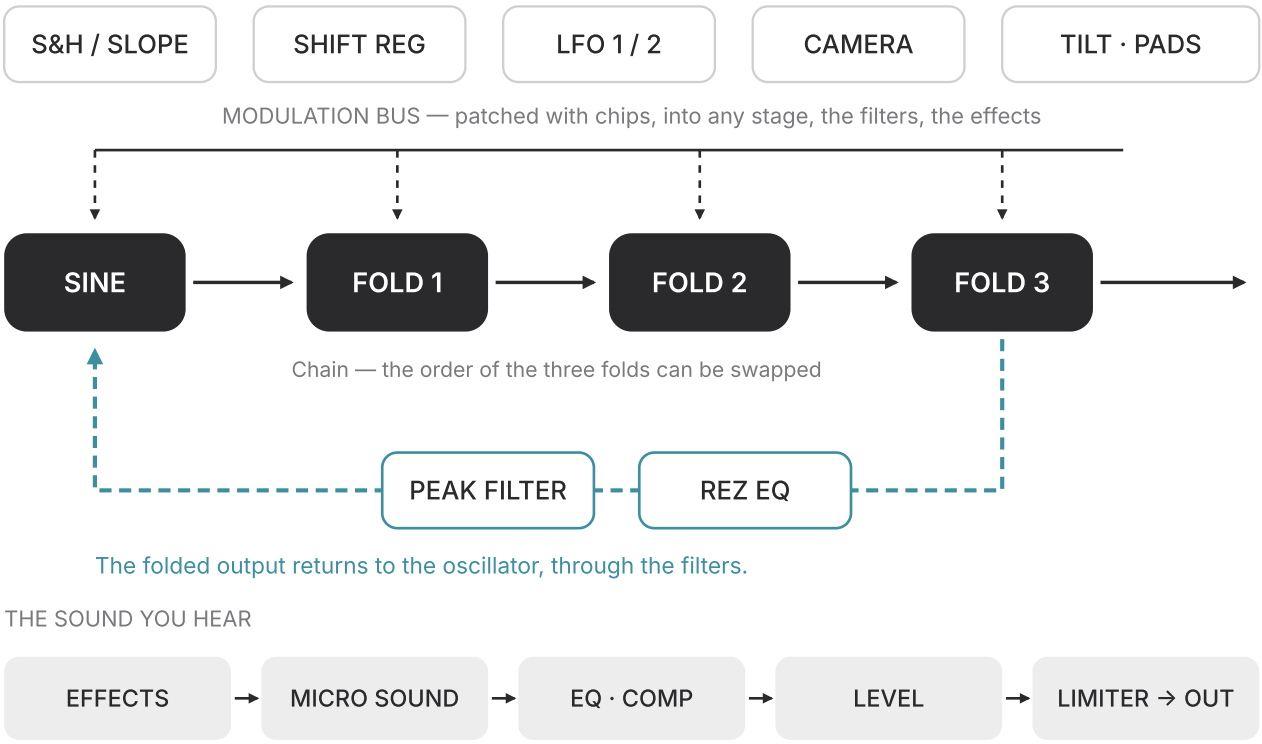
Sunnandæg is a **standalone app**, not an AUv3 plug-in. The camera, the tilt of the phone, the gravity ball and the microphone are part of the instrument itself, and a plug-in inside a host app cannot use them in the same way. It plays in the background and sends MIDI to other apps (section 14).

That said, we know many of you want to use it in finished pieces — to add improvised colour and texture to a track. With that in mind, **an AUv3 version is under consideration**. News will come through the newsletter: buttondown.com/kodk

03 · THE SIGNAL PATH

One loop

A sine oscillator, three wave folders in series, and a return path from the end of the chain back to the oscillator. Above them, a bus of modulators that can be plugged into almost anything.



Three feedback lines (one per fold) can also mix in another fold, either filter, or white / pink noise — see section 06.

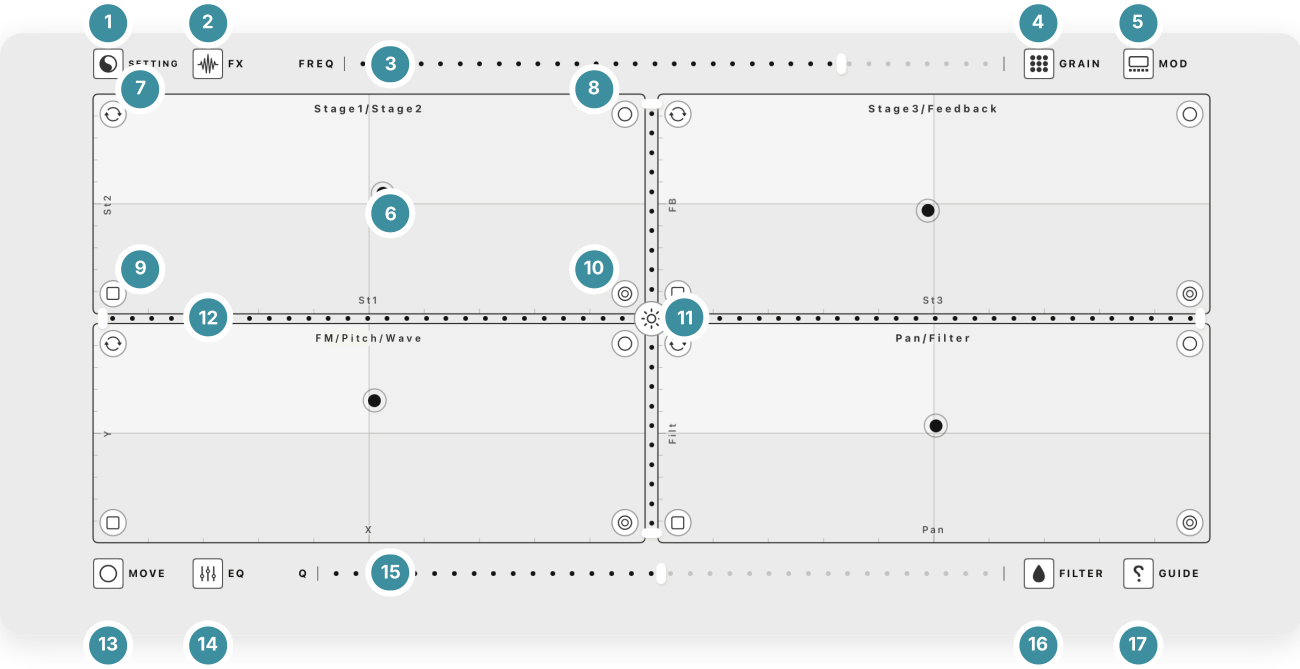
Going forward

The sine enters fold 1. Each fold bends the wave back on itself when it grows past a limit, adding harmonics instead of filtering them away. Pads 1 and 2 set how hard each stage folds.

Coming back

The end of the chain is sent back to the oscillator through the filters. Pad 2 sets how much returns. With a little, the tone thickens; with a lot, the folds start to call up more folding and the voice walks on its own.

Where your hands go



1. **SETTING** — System / Presets: engine, camera, tilt, ball, presets, random keys
2. **FX** — delay, reverb, phaser, chorus
3. **FREQ** — base pitch of the oscillator
4. **GRAIN** — Micro Sound
5. **MOD** — S&H / SR / Feedback Network, the patchbay
6. **The four XY pads**
7. **Swap** — exchange X and Y (by chance)
8. **Gesture recorder** for that pad
9. **Camera drive** for that pad
10. **Gravity ball** for that pad
11. **Camera mosaic** — all four pads at once
12. **The cross of sliders** — clocks and thresholds
13. **MOVE** — global recorder, all four pads, up to 12 s
14. **EQ** — EQ / Comp / Level / LFO
15. **Q** — sharpness of the peak filter
16. **FILTER** — peak filter / Rez EQ
17. **GUIDE** — the manual inside the app

Your first five minutes

1. Open the app. A plain sine plays. Move **FREQ** to find a register you like.
2. Drag on **pad 1**. The tone grows harmonics as the first two folds bite.
3. Raise **Y on pad 2** slowly. This is the return path — listen for the moment the voice starts to answer itself.
4. Tap the **double circle** on a pad and tilt the phone. The ball now plays that pad.
5. Open **MOD**, set ST's destination to Pitch, and switch the S&H on.
6. When it is somewhere good, **hold a preset slot** to keep it. When you are lost, press **Reset** to strip it back, or **RND** to find a new shape.

05 · THE FOUR PADS

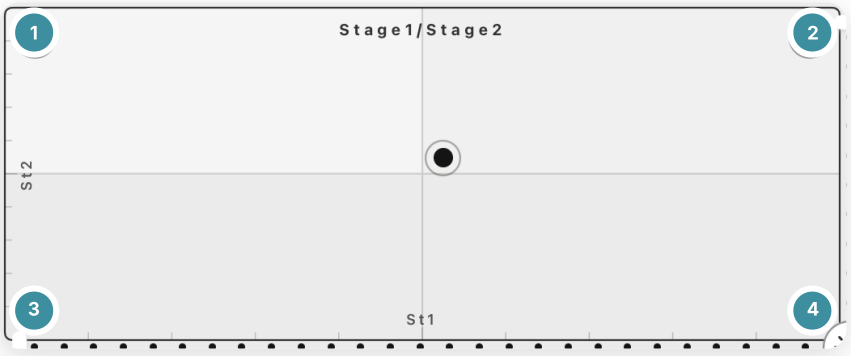
Four squares, eight hands

Each pad has an X and a Y, so the four pads give you eight values to play at once. Every one of them is also a sender: X1 to Y4 can be plugged into anything else.

PAD	X / Y	SENDS AS
1 · top left	Fold stage 1 / fold stage 2 — how hard each folds	X1 Y1
2 · top right	Fold stage 3 / the amount returned to the oscillator	X2 Y2
3 · bottom left	The oscillator's character — how the returning signal acts on it	X3 Y3
4 · bottom right	Stereo pan / filter	X4 Y4

The four quadrants inside a pad are separated by a very faint difference in tone — a quiet map of where you are.

Four ways to move a pointer



1. Swap X and Y (by chance)
2. Pad recorder
3. Camera drive
4. Gravity ball

Finger

Touch and drag. The default.

Camera drive (square)

The camera's colour moves the pointer: red against blue drives X, green against the others drives Y. The pad surface greys with the camera's brightness, and its border darkens with movement in that pad's own quarter of the picture.

Gravity ball (double circle)

The pointer becomes a ball. Tilt the phone and it rolls, bouncing off the walls; lay it flat and it settles. You can grab and throw it. With Gravity at zero it bounces forever.

Camera mosaic (sun, centre)

The picture is broken into a coarse mosaic across all four pads. Each pointer is pulled toward one cell — the one with the most movement, or the brightest or darkest (set in the Camera row).

Swap

Exchanges X and Y on that pad — but only sometimes; the outcome is random. When a swap really happens, the quadrant tones shuffle, so you can see it.

Recorders

Tap to record your movement, tap again to loop it, tap once more to return to your hand. Double-tap erases. The pad recorder loops one pad; the global recorder loops all four, stopping itself at 12 seconds. Presets keep the global loop.

The cross of sliders around the pads holds four values, one per half: left is the shift register speed, right the S&H division, bottom the shift register threshold, top the S&H threshold. A half is coloured only while its function is on.

06 · SEND AND RECEIVE

Who sends, who receives

Every modulation in Sunnandæg is patched the same way. A receiver — any parameter with a row of small chips under it — chooses one sender by tapping its chip. There are no menus and no cables: the whole patch stays on the surface, and one glance across a panel tells you what is driving what.

Tapping a chip

Tap a chip to choose that sender. The lit chip is the one in use.

Tapping a lit chip

Many chips hold more than one sender. Tap the chip that is already lit and it steps to the next one inside it. The label always says where you are.

X1 → Y1

F1 → F2 → F3

ST → SL → SR → CP

L1 → L2 — nothing patched

The senders

CHIP	WHAT IT SENDS
—	Nothing. The receiver stays where its slider puts it.
F1 F2 F3	The audio output of each wave folder stage. Patching a stage into something that feeds it again is where the instrument starts to misbehave — in the good way.
X1 … Y4	The eight pad axes. Whatever moves the pad — finger, camera, ball, recorder — comes out here.
ST	Stepped: a sample-and-hold value, grabbed on each gate and held.
SL	Slope: the same input, chased smoothly.
SR	The shift register: values marching along in sequence.
CP	The coupler: a gate that stands while the smooth side is above the stepped one.
L1 L2	The two free-running LFOs.
CM	The camera — how much movement it is seeing.
TL	The tilt of the phone.

Some rows show only the chips that make sense there, so not every receiver offers every sender. The map on the next page shows which.

06 · SEND AND RECEIVE — THE MAP

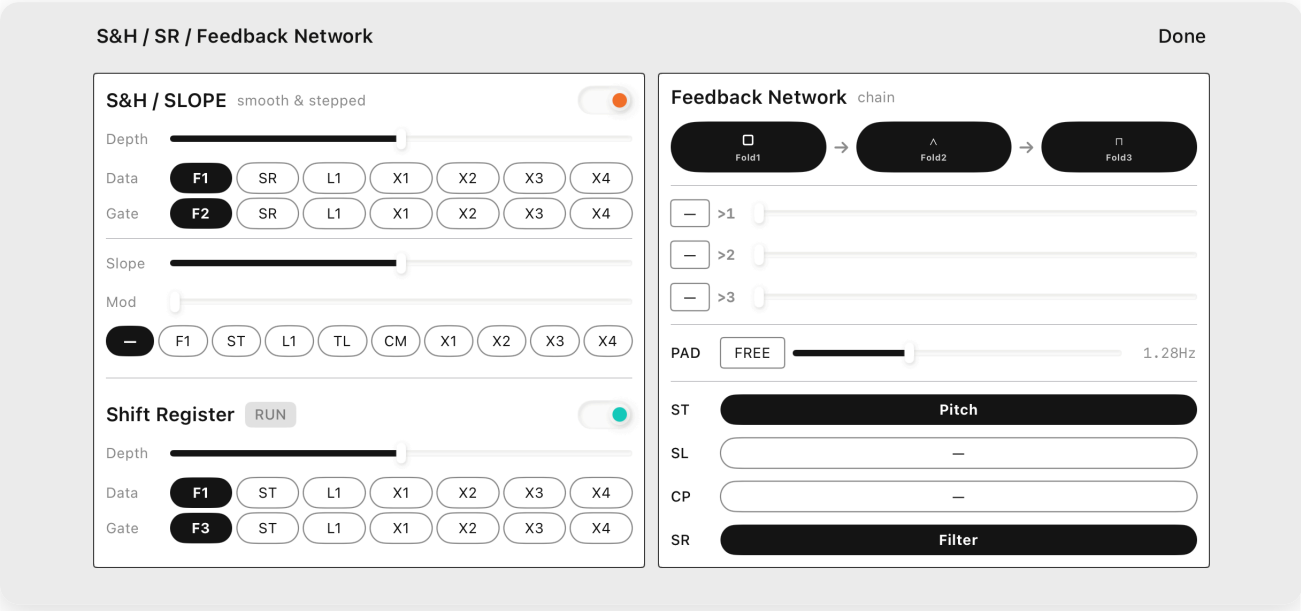
Which receiver listens to what

RECEIVER	F1–F3	ST GROUP	L1 / L2	TL	CM	X / Y
S&H · Data, Gate	●	SR	●			●
Slope · Mod	●	ST	●	●	●	●
Shift register · Data, Gate	●	ST	●			●
Peak filter · P1, P2	●	SR	●	●		●
Peak filter · Low, High, P-CV	●	ST	●	●		●
Peak filter · Q	●	ST	●	●	●	●
Rez EQ · Feedback, Delay	●	ST	●	●	●	●
Rez EQ · each of 8 bands	one chip per band — tap to step through the sources					
Effects · every parameter	pads, S&H, shift register, camera, tilt					
Micro Sound · Size, Pos, Pitch, Shape, Vol	each with its own amount and source					
LFO 1 / 2 · Fq In, FM In, Reset In	bends the speed, the waveform, or restarts the cycle					

“ST group” lists the first chip shown in that row; tap it again to reach the others in the group.

Two more ways things are sent — the S&H, slope, coupler and shift register can push themselves straight onto the instrument (Destination), and three feedback lines can return any fold, either filter or noise into each fold. Both live in the MOD panel — see section 07.

Stepped, smooth, and fed back MOD



S&H / Slope — stepped and smooth

One **Data** input, one **Gate**, two outputs. On each gate ST grabs the data and holds it; SL chases the same input continuously, **Slope** setting how fast, with its own Mod. CP, the coupler, stands high while the smooth side is above the stepped one. Fed by the folds, the S&H is clocked by the sound it shapes.

Shift register

Values march along in sequence — a pattern appears, mutates, dissolves. **Depth** sets how hard it lands; speed and threshold sit on the cross of sliders. RUN shows when it moves.

PAD — FREE / HOLD

HOLD samples the pads themselves: pointers and sound move only in steps, at 0.2 to 40 Hz, however smoothly you stroke. FREE lets them glide.

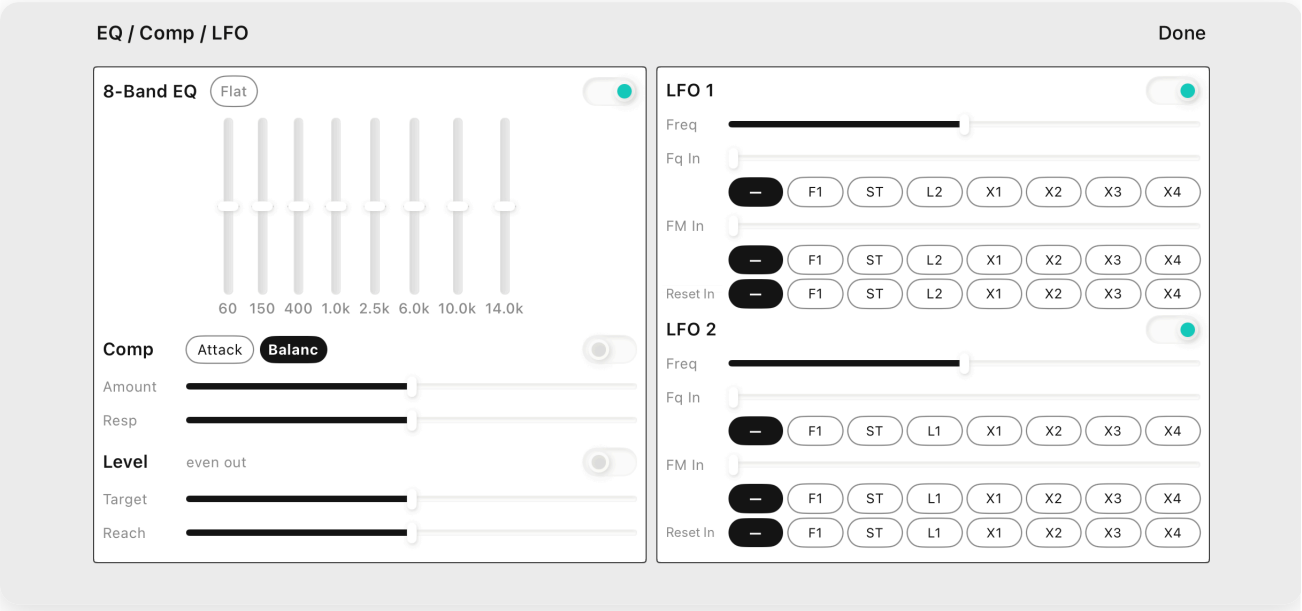
Feedback network

Chain reorders the three folds — tap a block to swap it with its neighbour. Below, three lines **>1 >2 >3** feed each fold: another fold F1–F3, the peak filter TW, the Rez EQ RE, or white / pink noise WN PN. A stage never feeds itself and no signal is patched twice. A little noise is colour, a lot is chaos.

Destination

ST SL CP SR each have one wide button — tap to step through **—, Pitch, Filter, Fold1–3, FB, Pan**. CP lands as a hard step at half depth, so it colours rather than pins.

Finishing, and two slow hands EQ



The output stage

8-Band EQ with its own switch; **Flat** returns every band to 0 dB.

Comp — Attack lets the front of each sound punch through and squeezes the sustain; Balanced evens the whole signal out. Amount and Resp.

Level — leans the volume toward one Target: loud passages held down, quiet ones lifted. Reach sets how far (up to 18 dB). It moves slowly on purpose and adds no delay. This instrument lives on swells of feedback, so leave it off unless you want the range narrowed.

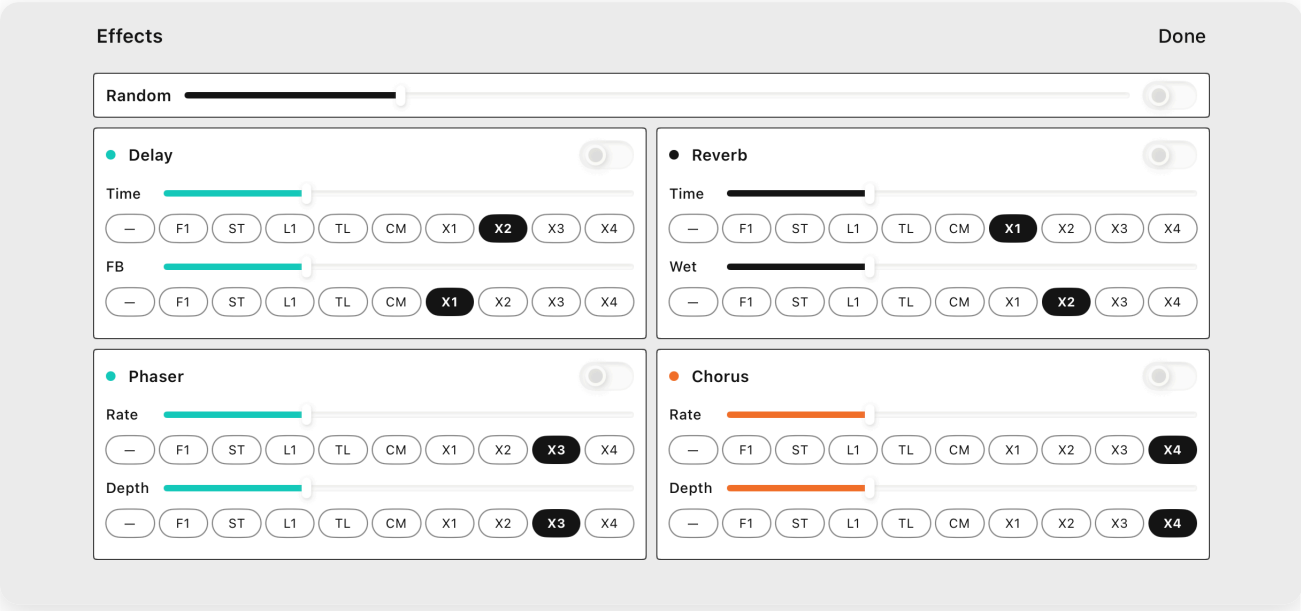
After all three, a transparent **limiter** guards the output — nothing to set.

LFO 1 and LFO 2

Two free-running sines. **Freq** sets the speed, **Fq In** modulates that speed, **FM In** adds to the phase so the waveform itself bends, and **Reset In** restarts the cycle whenever the chosen signal rises through its middle. Each LFO can be driven by the other.

They come out as L1 L2 everywhere else.

Four rooms for the sound FX



Delay — Time and FB (feedback). **Reverb** — Time and Wet. **Phaser** — Rate and Depth. **Chorus** — Rate and Depth.

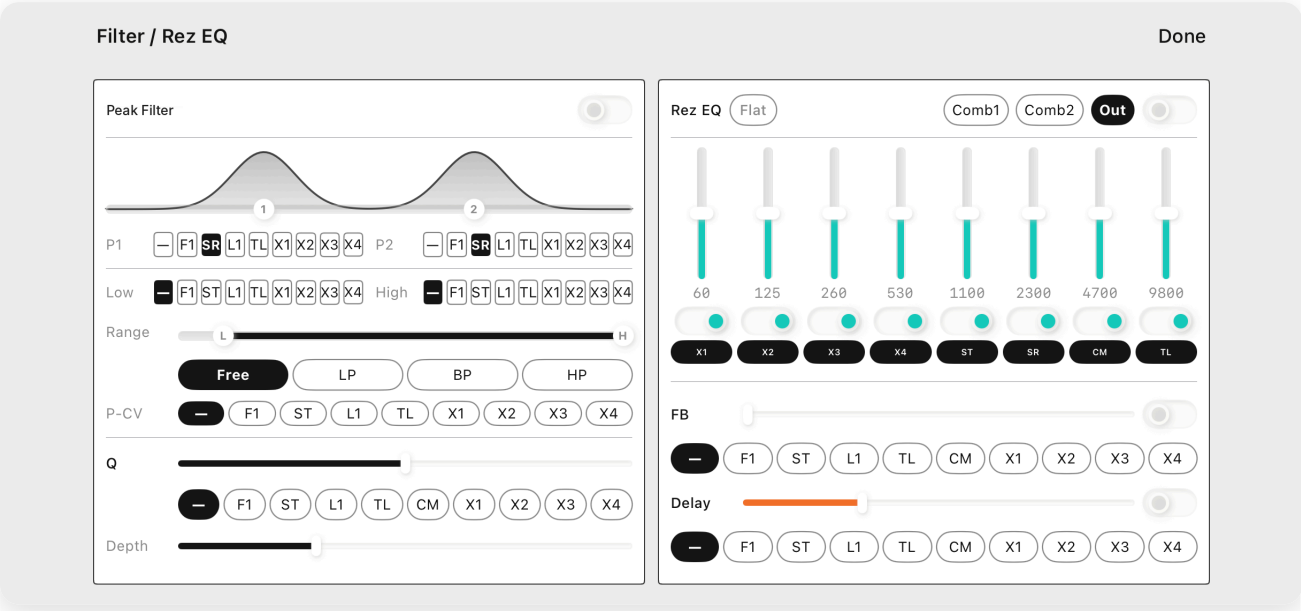
Each effect has its own switch, and every parameter has a chip row underneath: leave it on — and the slider decides; pick a pad, the S&H, the shift register, the camera or tilt and it moves on its own.

Random

With Random switched on, the active effect is reselected automatically every time your pad touches reach the count set by its slider — the room changes as you play.

The dot beside each name is the colour its sliders take on.

Two filters in the return path FILTER



Peak filter

A resonant filter with two peaks. **Position** places them; **P1** and **P2** choose what drives each one, and **P-CV** moves both together while keeping their spacing. **Low** and **High** pick what drives each end of **Range**, which limits how far the peaks travel. Free / LP / BP / HP narrows what passes without touching the resonance. Push **Q** high and it begins to sing on its own; **Depth** sets how far its modulation reaches.

Rez EQ

Eight fixed resonant bands, 60 Hz to 9.8 kHz, each with a level fader, a switch and its own source chip; **Flat** resets them. Output: odd bands (Comb1), even bands (Comb2) or all (Out). **FB** sends the result back into the bands, optionally through a short **Delay** (1–50 ms) for metallic, comb-like colour.

The sound, made of particles GRAIN



Source — **INT** records the instrument itself while you hold; **EXT** records the room (the instrument goes quiet meanwhile); **FILE** loads the first 30 s of an audio file. Play / stop run the material at its own speed.

Trim — handles S and E; only what lies between is reachable.

VCA — FORM lets the grain's envelope open it; AUD multiplies the grains by the live waveform, so they become that sound made of particles. **Vol** has its own modulation.

The **trapezoid** is the grain envelope — narrow for sharp particles, wide for soft. **Pos** slides it, **Shp** widens it. **Voices**: one to four.

Gate picks what throws the grains: F1–F3 cross far more often than OUT, so grains thicken deeper in the folds.

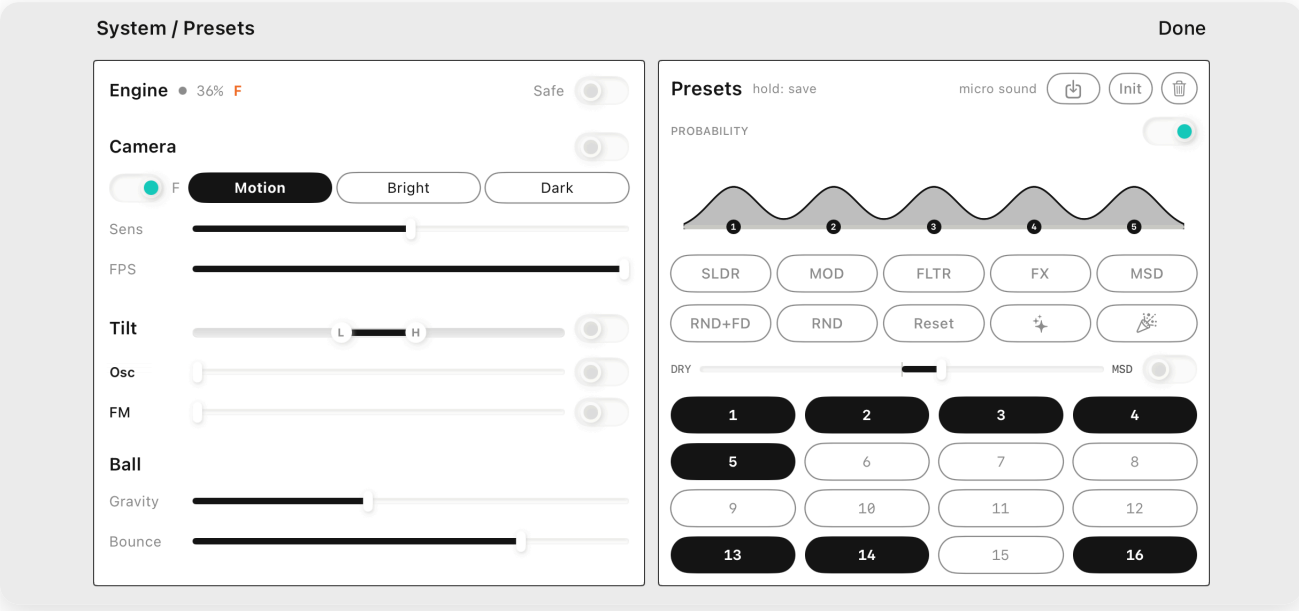
Thresh is a proportion of the current swing: at the bottom it fires once per cycle — the grain rate becomes your pitch; at the top only the biggest peaks pass.

Size 1–300 ms · **Pos** · **Pitch** ±2 octaves, each with amount and source. **Scan** moves the head (centre freezes), **Spread** scatters, **Jitter** detunes, **Width** spreads in stereo.

The dry / grain balance (MSD) sits in the SETTING panel.

12 · SETTING — SYSTEM AND PRESETS

The machine, and your places SETTING



Engine — load (amber over ~70%, crackles near 100%) and a letter when the phone warms: F, S, C. **Safe** lowers the camera rate and widens the audio buffer when it is hot.

Camera — master switch, front (F), what the mosaic pointer follows (Motion, Bright, Dark), **Sens**, **FPS** (lower runs cooler).

Tilt — L / H set how far the pitch travels at full tilt; two more rows send tilt into the oscillator. A row switched off is cut completely.

Ball — Gravity and Bounce.

Presets — sixteen slots for the whole state, gesture loop included. **Hold** to save, **tap** to recall; grain material under ten seconds is kept too. Beside the heading: the Micro Sound load key, **Init**, and the trash.

Probability — five hills decide where random pitches land: crowd them together or spread them out.

DRY — MSD — the balance between the plain sound and the grains, with the switch that turns grains on. No random key touches it.

The random keys are on the next page.

13 · RANDOM KEYS AND GESTURES

Losing your place, on purpose



By corner

SLDR	the pads and the front sliders
MOD	S&H, shift register, LFOs, feedback network
FLTR	peak filter and Rez EQ — one filter always sounds
FX	which effects are on and how hard
MSD	how grains are cut and linked — never the material

Whole instrument

RND+FD	fresh values for the playing side
RND	nudges the sliders, rewires every modulation — recognisable, new shape
Reset	strips: effects off, feedback to zero; your patch stays
✦	one push — an eighth of RND+FD, pitch and clocks left alone

Hold to glide

Wiring and switches change at once; everything continuous travels to its new value over up to five seconds. Lift your finger and it arrives — a tap is instant, a long press a slow morph. None of these touch the output stage, Safe, the drive modes or your presets.

The party popper — a gesture

Not random at all. It plays one of fourteen gestures — a reverb tail with the source cut away, a ringing delay, an accelerating drill, a flutter, a held breath, a chopped grid, a phaser-and-chorus wash, a pulse, a swell, a strobe, a frozen loop, everything at once, a roll into a buzz, a torn stutter — then puts everything back exactly as it was. Its name shows on the key; tap for its own length, hold to make it last.

14 · MIDI AND CREDITS

Playing other things

Sunnandæg does not receive notes, but it can send. Its movement can drive other apps and instruments.

Where MIDI goes

- A **virtual MIDI source** for other apps on the same iPhone (AUM and similar).
- **Network MIDI** to a Mac on the same Wi-Fi. On the Mac, open Audio MIDI Setup → MIDI Studio → Network, find the iPhone under Directory, and Connect.
- Any **connected MIDI device**, over USB or Bluetooth.

What it sends

- **CC out** — the eight pad axes (X1 to Y4) as control changes, on the channel and CC numbers you set. When CC out is switched on, all eight current values are sent at once.
- **Note out** — the shift register (or the S&H) becomes a melody, quantised to a key and scale you choose. Velocity follows the actual output level.

Background and lock screen

The instrument keeps playing when you leave the app, and can be stopped from the lock screen and Control Centre.

Where it comes from

Two ideas from modular synthesis stand behind this instrument. One is the west-coast wave multiplier — folding a plain sine until it grows a spectrum of its own, rather than filtering harmonics away. The other is the chaotic, self-patching approach of the shift-register tradition, where a handful of stepped values turns into music that never quite repeats. Nothing here is a copy of those designs — no circuit or code was taken from them — but the way this instrument listens to itself owes them a great deal.

15 · VERSIONS

Version 1.2.0

CURRENT

1.2.0

iPhone and iPad, landscape · iOS 18.0 or later. This guide describes this version.

What’s new in 1.2.0

iPad	Sunnandæg runs on iPad, in landscape and full screen. The whole instrument fills the display, so pads are taller and every button and label is larger.
Panels on iPad	Every panel opens full screen, keeping exactly the same layout as on iPhone, scaled up to fill the display.
Taller pads	The height of the pads is now measured from the screen: any spare vertical space goes to the pads.
Camera surface	With the camera on, the pads return to their original opaque, recessed surface with a simple black pointer. The mosaic now shows only as thin bands at the top and bottom, which react as the frame.

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